



Subject Vision and Intent: Science

Our Vision

At King's Cross Academy, we want every child to see themselves as a scientist. We believe that Science enables children to understand the world around them, ask questions about what they observe and develop the knowledge and ways of thinking that allow them to seek explanations.

Our ambition is for children to leave KCA curious about the natural and physical world and equipped with the scientific knowledge, vocabulary and thinking they need to make sense of it. We want children to experience the excitement of discovering something new while understanding that scientific knowledge is built through observation, investigation, evidence and the willingness to reconsider ideas when new evidence emerges.

Our Science curriculum reflects our Academy vision, *Love Learning Together*. Science encourages children to share ideas, question assumptions, investigate collaboratively and learn from evidence. It also develops our KCA values: the courage to ask questions and test ideas, the integrity to respond honestly to evidence, the endurance to investigate when answers are not immediately apparent and the aspiration to understand more about the world.

Our Intent

Our Science curriculum is ambitious, inclusive and carefully sequenced. It identifies the essential knowledge, vocabulary, concepts and scientific skills we want children to develop from the Early Years to Year 6, enabling them to build increasingly rich and connected knowledge across biology, chemistry and physics.

Scientific knowledge is deliberately built over time. New learning connects with and develops what children have encountered before, allowing important concepts to be revisited with increasing depth and complexity. Children learn to make connections between different areas of science and use what they already know to understand new phenomena and explain the world around them.

Alongside developing this substantive knowledge, children learn to work scientifically. They ask questions, make observations, classify and compare, measure, investigate, identify patterns, gather and analyse evidence, draw conclusions and communicate their findings. As

they progress through the school, children become increasingly independent in deciding how questions might be investigated and how evidence can be used to support or challenge an explanation.

Practical science is an important part of our curriculum. Children have opportunities to observe phenomena, handle materials and equipment, investigate questions and apply their scientific knowledge. Practical experiences are carefully chosen to deepen understanding, develop scientific skills and enable children to experience how scientific knowledge can be explored and tested.

We want children to understand the importance of evidence. They learn that scientific explanations should be supported by what we can observe, measure and investigate, and that results do not always support our original ideas. Children are encouraged to question, reconsider and refine their thinking in response to what the evidence tells them.

Scientific vocabulary is explicitly taught and revisited so that children develop the language needed to describe observations, explain concepts, communicate findings and reason scientifically with increasing precision.

Our curriculum also helps children understand the relevance of science to everyday life and the wider world. Through real-world contexts, children encounter how scientific knowledge contributes to medicine, technology, engineering, the environment and our understanding of the universe. They learn about scientists from different backgrounds and the contribution scientific discovery continues to make to our lives.

Every Child

We have high expectations for every child. Children with SEND, children who are disadvantaged and children who are developing English as an additional language access the same ambitious Science curriculum.

Teachers identify the essential knowledge, vocabulary and scientific concepts that all children need to understand and adapt teaching so that barriers are reduced without lowering expectations. Explicit teaching, carefully chosen practical experiences, visual representations, models, structured discussion, vocabulary support and opportunities to revisit important knowledge enable children to develop secure scientific understanding.

We recognise that children's experiences of science beyond school can vary considerably. Our curriculum therefore ensures that every child has opportunities to investigate scientifically, handle equipment, encounter the natural world and understand how science operates beyond the classroom.

We want every child to experience the satisfaction of asking a question and finding something out for themselves. Children are encouraged to recognise that curiosity, careful observation and thoughtful questioning are important characteristics of scientists and that science is a discipline in which everyone can participate.

Science Beyond the Curriculum

Our location provides exceptional opportunities to connect children's scientific learning with the world beyond school. King's Cross is home to an extraordinary scientific and

technological community, allowing children to see that the science they encounter in the classroom is a living discipline taking place around them.

Through our links with organisations including the Francis Crick Institute, children have opportunities to encounter scientists, scientific research and real-world applications of the knowledge they study. These experiences help children understand what scientists actually do and broaden their awareness of the range of careers and possibilities associated with science.

Our local environment also provides opportunities for scientific observation and investigation. Experiences beyond the classroom allow children to apply their knowledge, ask questions about the world immediately around them and recognise that opportunities for scientific enquiry exist everywhere.

By the end of Year 6, our aim is not simply that children remember a collection of scientific facts. We want them to have developed connected scientific knowledge that they can draw upon to explain phenomena, ask meaningful questions, interpret evidence and develop increasingly sophisticated explanations of the world.

Above all, we want children to leave KCA as curious, knowledgeable and confident scientists: children who are excited by questions, guided by evidence and equipped with the foundations to continue exploring and understanding the world around them.